

23SLA2

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Problem

Let $\mathbb{R}$ be the set of real numbers. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that

$$f(x+y)f(x-y) \geq f(x)^2 - f(y)^2$$

for every $x, y \in \mathbb{R}$. Assume that the inequality is strict for some $x_0, y_0 \in \mathbb{R}$. Prove that either $f(x) \geq 0$ for every $x \in \mathbb{R}$ or $f(x) \leq 0$ for every $x \in \mathbb{R}$.

Solution. Let $P(x, y)$ be the assertion.

$P(x, x) \implies f(2x)f(0) \geq 0$. If $f(0) \neq 0$ then $f(2x)$ has the same sign as $f(0)$ which is constant.

Now let $f(0) = 0$.

$$P(x, y) + P(y, x) \implies f(x+y)[f(x-y) + f(y-x)] \geq 0.$$

$$P(x, -x) \implies f(-x)^2 \geq f(x)^2$$

$$P(-x, x) \implies f(x)^2 \geq f(-x)^2$$

$$\implies f(x)^2 = f(-x)^2.$$

$$P(x_0, y_0) + P(y_0, x_0) \implies f(x_0 + y_0)[f(x_0 - y_0) + f(y_0 - x_0)] > 0$$

If $f(x_0 - y_0) = -f(y_0 - x_0)$ then,

$$P(x_0, y_0) + P(y_0, x_0) \implies f(x_0 + y_0)[f(x_0 - y_0) + f(y_0 - x_0)] = 0 \text{ which is a contradiction.}$$

Let $x_0 - y_0 = b$.

So, $f(b) = f(-b)$.

Observe that $f(a)(f(c) + f(-c)) \geq 0$ for all pairs (a, c) since for any given a, c , its corresponding x, y can be constructed.

So, Let $Q(a, c)$ be the assertion that corresponds to this.

$Q(a, b) \implies 2f(a)f(b) \geq 0$. Since $f(b)$ is fixed and $f(a)$ and $f(b)$ have the same sign, $f(x)$ has a fixed sign, so $f(x) \geq 0$ for all x , or, $f(x) \leq 0$ for all x .